

Developing Steady-State Cornering CFD Simulations for Use in FSAE

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Work done in conjunction with
Siemens and Queen's Formula SAE

1 Introduction

The developed simulation was designed to meet the goals of modelling our car in a steady-state corner, primarily tackling the skid pad event. The base simulation can take on any desired corner radius, initially set up for a left-hand turn. This simulation makes extensive use of global parameters and coordinate systems to implement the cornering methodology. As a note, this simulation can be implemented on any general meshing framework of your choosing.

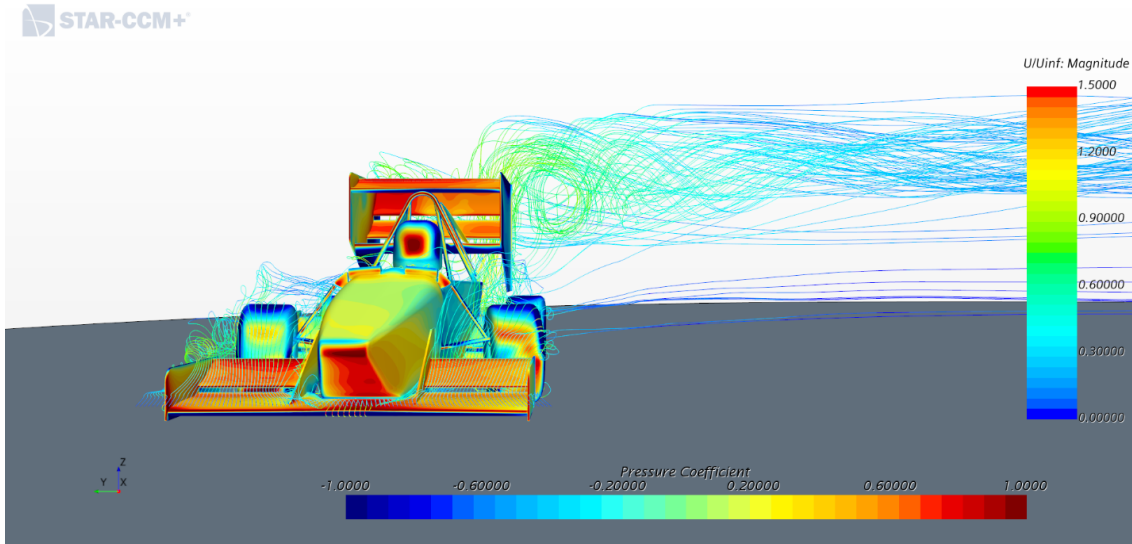


Figure 1: Streamlines of a cornering simulation for a car yaw angle of zero degrees on a skid pad-sized corner.

The corner is modelled as a pie-shaped domain through which the vehicle rotates, in a moving reference frame (MRF). An inlet velocity of zero is specified, as the fluid motion is generated by the motion of the vehicle.

This article will instruct teams on how to implement this framework into existing simulations, rather than act as a standalone simulation file. Queen’s Formula uses a mesh file and a run file to help reduce overall simulation sizes. This architecture may not be preferable for some teams, so feel free to adapt this framework into your own team’s methodology. This simulation has been of value in developing aerodynamic yaw moment diagrams for a constant steering angle. It can provide insight on the stability characteristics of a team’s package, as well as the car’s sensitivity to yaw in a corner.

2 Before You Start

This methodology has been implemented in various versions of STAR-CCM+ from 2019.1.1 to 2020.1.1. Global Parameters have been available prior to version 13, so users of older versions can incorporate these methods.

3 Creating the Domain

The domain of the corner was created in STAR-CCM+ CAD modeler, such that design parameters can be created to update the domain sizing quickly for desired corner radii. The following design parameters were created in the CAD modeler:

- Radius (R): The desired corner radius in metres
- Domain Height: The height of the domain in metres, recommended setting is 16 m
- Wheelbase: Car’s wheelbase in metres

- CGX: Percentage of the car's weight on rear tires, defines Centre of Rotation (CoR)
- CG: $\{CGX\} * \{Wheelbase\}$
- Outer Radius (OR): Outer wall of the domain in metres
- InletAngleWheelbase (IAW): Arc length definition from CoG to inlet face in number of wheelbases
- OutletAngleWheelbase (OAW): Arc length definition from CoG to outlet face in number of wheelbases

For an 8.375-metre corner (FSAE Skidpad), the recommended OR, IAW and OAW are 50 m, 9 and 18, respectively. To start the domain, an initial sketch is made on the XY plane to create the following:

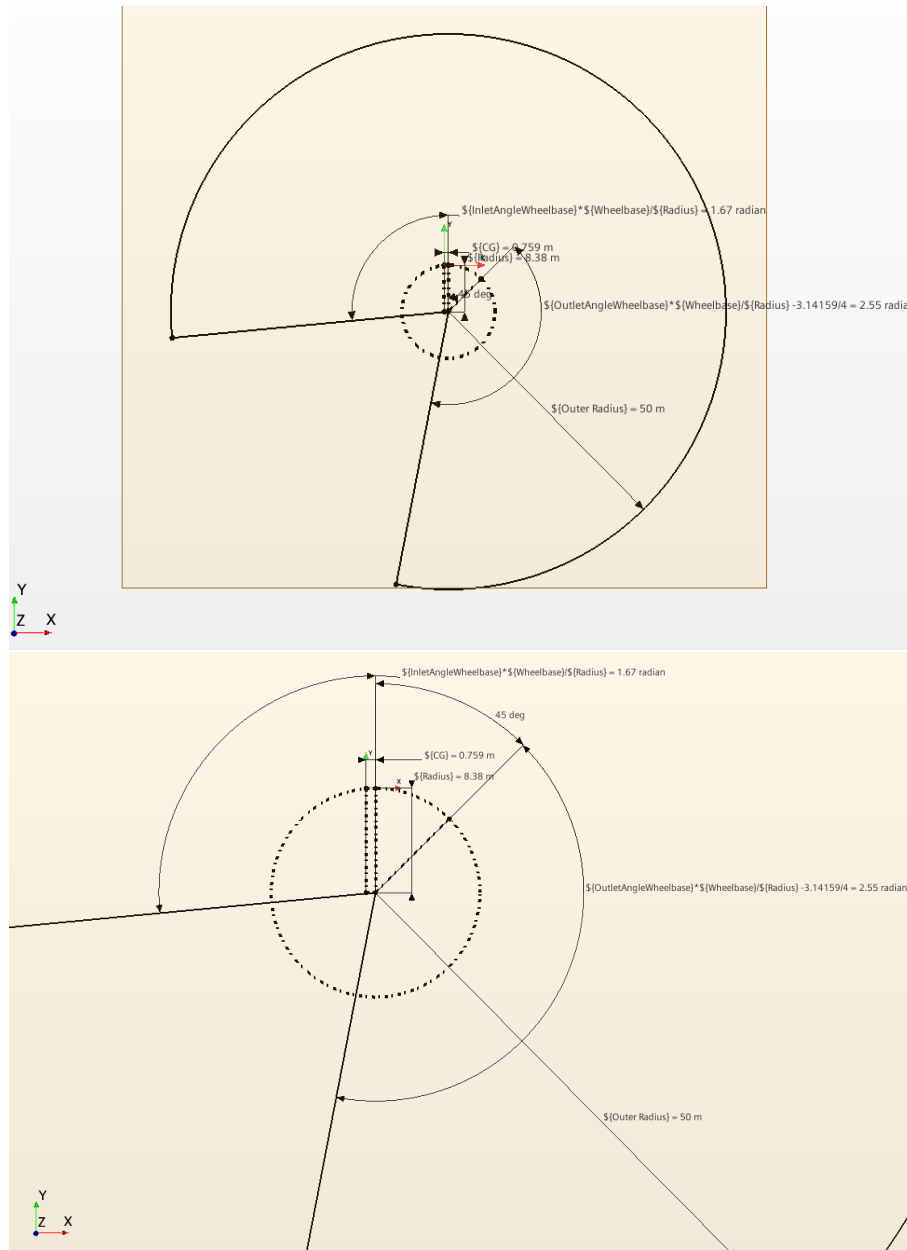


Figure 2: Sketch of the base domain. All dimensions with a \$ are based on parameters. The lab coordinate system is found in the centre of the picture, The close up of the center of the domain is found at the coordinates ($\{CG\} - \{Radius\} \ 0$) in the global coordinate system.

This dimensioning scheme assumes that the lab coordinate system is centred on the plane on which the front axle of the car (centre-to-centre) sits. Accordingly, the centre of the do-

main would be $(\text{cgx} * \text{wheelbase} - \text{Radius } 0)$, where **Radius** is the corner radius. The inlet face of the domain is located based on a variable arc length formula, defined by $\{\text{InletAngleWheelbase}\} * \{\text{Wheelbase}\} / \{\text{Radius}\}$. The outlet is implemented by defining an angle between the outlet face and a line that is 45° from the CoG with the formula $\{\text{OutletAngleWheelbase}\} * \{\text{Wheelbase}\} / \{\text{Radius}\} - 3.14159/4$. The angle is implemented within the 45° from the CoG to ensure that the geometry does not fail for larger corner radii. The 3.14159/4 in the outlet length definition accounts for this. This has been tested for corner radii from 8.375 m to 32.5 m; these changes are made by varying the value of the Radius design parameter. Wider corners can be used, but the user must ensure that the value for OutletAngleWheelbase is increased to ensure the angle does not become negative. Should this value become too large for small corners (28 for a radius of 8.375 m), it will cause the inlet and outlet to overlap, also creating a failed sketch. This domain sketch is then extruded by 16 m, based on best practices.

For the creation of the wake refinements, three levels of refinement were used. They are also created in the CAD-modeller, using a lofted volume and making use of Boolean splits to create each refinement zone. The width at the start and end of the wake refinement is set to 2.5 m and 6 m respectively, with a linear distribution in between. The height goes from 2.5 m to 4.5 m, also with a linear distribution in between. The start of the wake refinement is an angle of $0.45 * \{\text{Wheelbase}\} / \{\text{Radius}\}$ from the CoG of the vehicle, and the outlet is coplaner with the outlet face of the domain. The first and second wake refinement blocks have angles of $4 * \{\text{Wheelbase}\} / \{\text{Radius}\}$ and $4.5 * \{\text{Wheelbase}\} / \{\text{Radius}\}$ respectively. The final refinement block has two internal profiles to ensure the geometry does not skew with a higher OutletAngleWheelbase value. The Start Contact Type and End Contact Type of the loft are both set to Contact Normal. The following table defines the angle, width and height for each section. Parameters have been abbreviated to save space. The prior section for Section 1 will be the line from the CoR to the CoG.

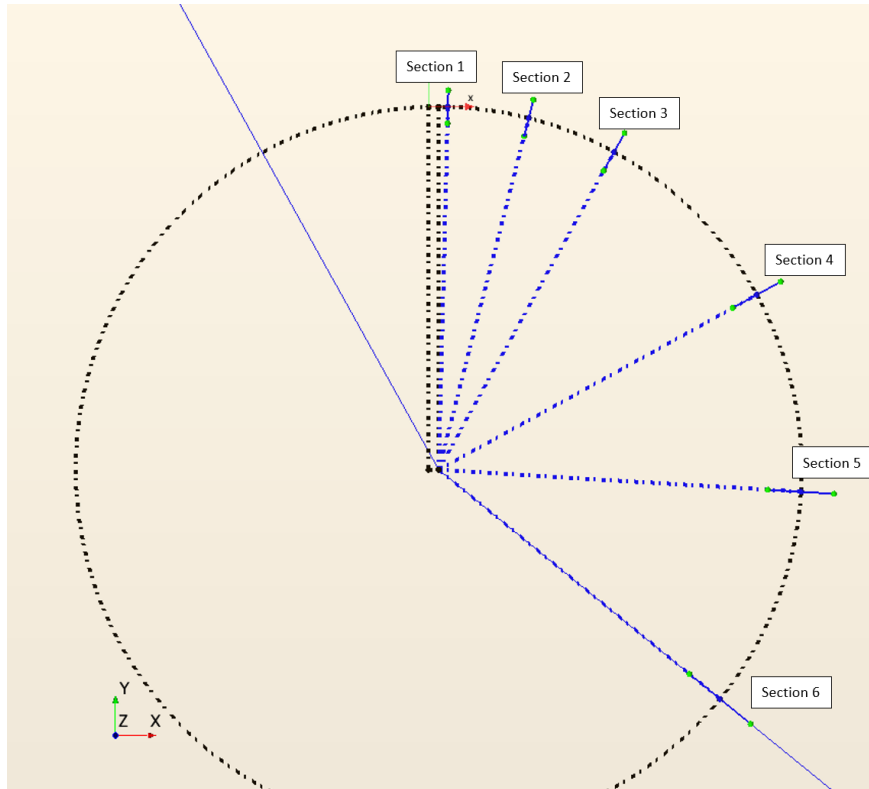


Figure 3: Sections defined for creating the wake refinements

Section	Width Definition [m]	Height Definition [m]	Angle From Prior Section
1	2.5	2	$0.45 * WB/R$
2	$2.5 + 3.5/(OAW - 0.45) * 4$	$2 + 2.5/(OAW - 0.45) * 4$	$4 * WB/R$
3	$2.5 + 3.5/(OAW - 0.45) * 8.5$	$2 + 2.5/(OAW - 0.45) * 8.5$	$4 * WB/R$
4	$2.5 + 3.5/(OAW - 0.45) * (8.5 + 0.25 * OAW)$	$2 + 2.5/(OAW - 0.45) * (8.5 + 0.25 * OAW)$	$4 * WB/R$
5	$2.5 + 3.5/(OAW - 0.45) * (8.5 + 0.5 * OAW) * 4$	$2 + 2.5/(OAW - 0.45) * (8.5 + 0.5 * OAW)$	$0.25 * OAW * WB/R$
6	6	4.5	$(0.5 * OAW - 8.95) * WB/R$

Table 1: Definitions for the section sketches when defining the wake refinements

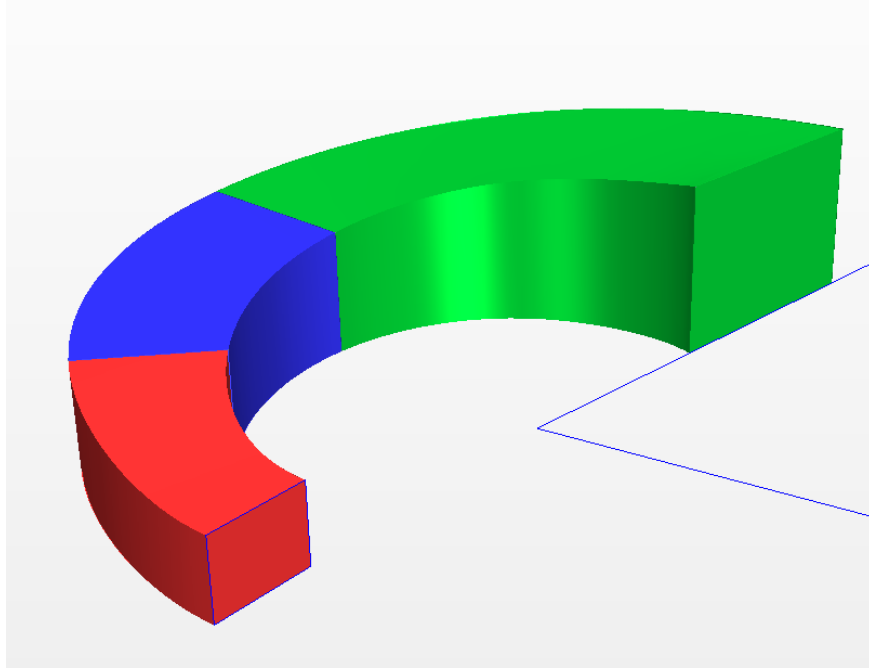


Figure 4: Final Refinement Blocks

4 Tire Orientation Formulation

To change the axis of rotation for each of the tires, a macro was recorded to determine what commands are necessary to change the basis vectors for the x- and y-axes, as well as the origin for each tire system. Separate mathematical functions were developed to determine these values in 3D space based on the Queen's Formula CAD techniques. As with the domain CAD, the assumption in this macro is that the front axle lies along the $x = 0$ plane for a 0° yaw car. The final matrices represent the x and y axis, where the positive z axis is the rotation axis for each tire, following the right-hand rule. To use the macros, simply modify the parameters at the top of the file, as per the comment's explanation of parameter input. When choosing the desired steering angle, typically the geometric angle to meet the desired turn radius is used given by $\delta = \arctan\left(\frac{l}{R}\right)$, where δ is the steering angle in degrees, with a left turn being positive, l is the car's wheelbase and R is the corner radius. Ackermann or anti-Ackermann geometries is ignored, but the macro does allow for different steer angles on each front tire.

STAR-CCM+ requires the definition of two basis vectors and an origin when creating a new coordinate system.

The x and y vectors for the front right wheel's coordinates can then be defined by the matrix:

$$\begin{bmatrix} \cos(\gamma)\cos(\phi) - \sin(\gamma)\cos(\theta)\sin(\phi) & \sin(\gamma)\cos(\phi) + \cos(\gamma)\cos(\theta)\sin(\phi) & \sin(\theta)\sin(\phi) \\ \sin(\gamma)\sin(\theta) & -\cos(\gamma)\sin(\theta) & \cos(\theta) \end{bmatrix}$$

The formulation of this matrix and the definition of each parameter are defined in a separate document. This is a step-by-step process, where each tire/vehicle parameter (yaw, toe, camber, steering) is defined as a series of rotation matrices. It also includes the assumptions and sign conventions used for each parameter, and how the above matrix needs to be modified for each tire. The document outlines the derivation of the origin for each tire too, which can be defined below for the front right tire by:

$$\begin{bmatrix} cg_x l - h_F \cos(\Phi_{FR}) & h_F \sin(\Phi_{FR}) & R_{FR} \end{bmatrix}$$

The attached Change_Parameter_Corner.java macro uses these same constituent equations to calculate the origins for each tire. Alternatively, these could be defined as separate vector parameters in STAR-CCM+, with all tire parameters being scalar parameters in simulation. This is an acceptable - but not suggested - method, as it can make the parameter workflow quite cumbersome.

5 Creating Global Parameters, Coordinate Systems and Reference Frames

For an easier implementation of the cornering methodology, parameters were used extensively. To create a parameter, navigate to the node Tools > Parameters. Right-click to create scalars or vectors. Only scalars are required for this simulation. To create formulas for the definition of a scalar, the convention $\{\text{parameter}\}$ is used for scalars and $\$\{\text{parameter}\}$ for vectors. The following parameters were created for the simulation:

- g, acceleration due to gravity, value: 9.81 m/s
- a_c, the desired lateral acceleration, definition: (desired accel in g's) * $\{g\}$, e.g. 1.1* $\{g\}$
- cgx, as defined in the orientation formulation, e.g. 0.49
- wheelbase, as above for the domain definitions, e.g. 1.55 m
- Radius, corner radius, e.g. 8.375 m
- XX_Rad, distance from center of rotation to each tire, defined in orientation formulations, eg. 8.0152
- Tire_Radius, tire radius in metres, e.g. 0.2286 (high profile 10" Hoosier's)
- VMag, magnitude of tangential velocity at the CoG derived from $a_c = v^2/r$, definition: $\sqrt{\{a_c\} * \{Radius\}}$
- OmegaXX, rotational speed for XX (e.g. FL) Tire, definition: $\sqrt{\{a_c\} * \{XX_Rad\}} / \{Tire_Radius\}$
- Domain_Rot, rotational speed of the car at the CoG, definition: $\{VMag\} / \{Radius\}$

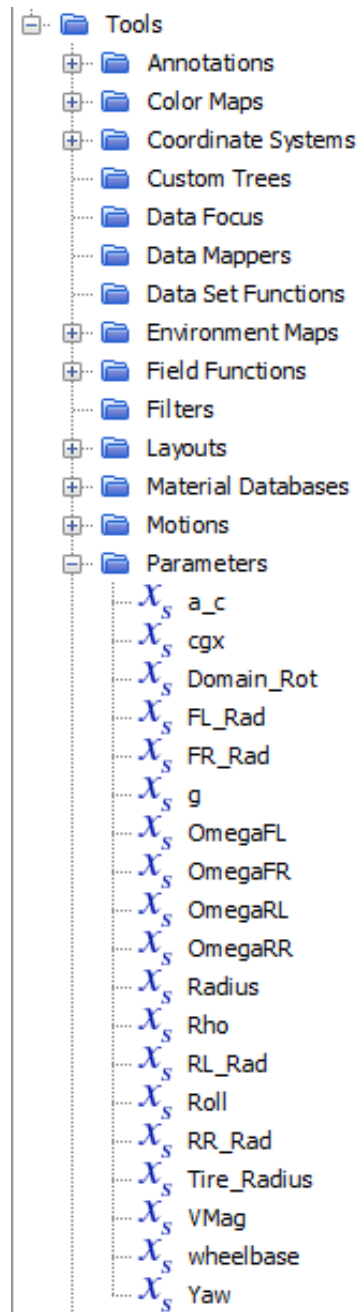


Figure 5: Location of the Parameters node in STAR-CCM+ workflow. X_s denotes a scalar parameter.

The VMag parameter is especially useful in any coefficient report that depends on the freestream velocity. With any change to the corner radius or target lateral acceleration, the normalization factor will automatically update. This parameter is also useful in the Field Functions node to normalize the pressure coefficient correctly, and to create a custom normalized velocity vector field defined by $\frac{\text{Velocity}}{\text{VMag}}$.

A Cartesian coordinate system must then be created for each wheel and the centre of rotation. This can be found under the node Tools > Coordinate Systems > Laboratory > Local. Right clicking on Local will allow new Cartesian coordinate systems to be created.

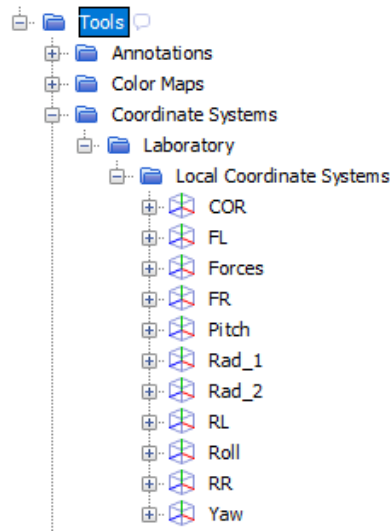


Figure 6: Location of the coordinate system node, along with necessary coordinate systems. Systems shown here that are not listed above are used for QFSAE-specific tasks.

The definition of the tire coordinate systems can be implemented using the attached macro. For the COR system, the origin is defined by $(\{cgx\} * \{wheelbase\}, -\{Radius\}, 0)$; leave the axis directions to be the same as the lab coordinates. This also assumes that the positive x-axis points towards the rear of the car. If teams have the x-axis reversed, the x definition must be made negative. Doing so centres the point of rotation to be concentric with the domain.

The final step before modifying boundary conditions is creating the rotating reference frame. This can be found under the node Tools > Reference Frames. Right clicking this allows for a New Rotating reference frame to be created. The axis of rotation is set to be the positive z-axis (or whichever points up in your simulation), with a rotation rate of $\{Domain_Rot\}$. The coordinate system should be set to Laboratory –> COR.

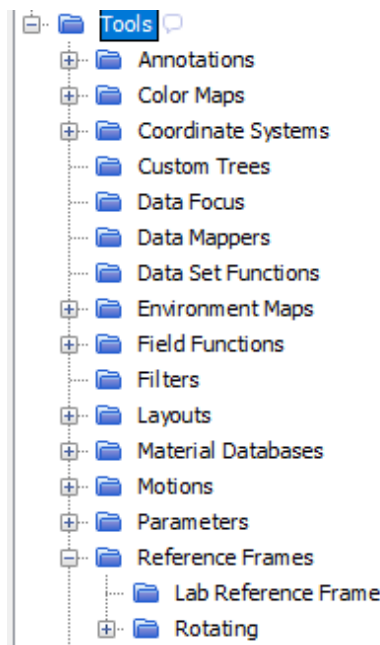


Figure 7: Location of the reference frame node.

6 Boundary Condition Modification

This simulation uses the Two-Layer K-Omega SST turbulence model but has also been successfully tested with the Two-Layer RKE turbulence model. Modelling has also been tested with boundary layer resolved and modelled meshing methods. Boundary layer resolved considerations are more accurate, however both techniques can be used depending on computational resources available.

The first step in applying the rotating frame, is to set the motion of all regions to be stationary in the rotating frame. This applies to not only the external flow region, but any radiator or fan regions (for the Q20 car there are two radiators and no fans). STAR-CCM+ now assumes all bodies in the simulation to be rotating, including

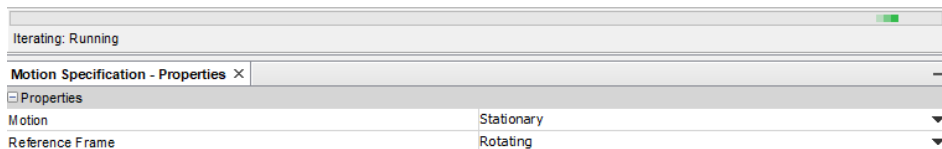
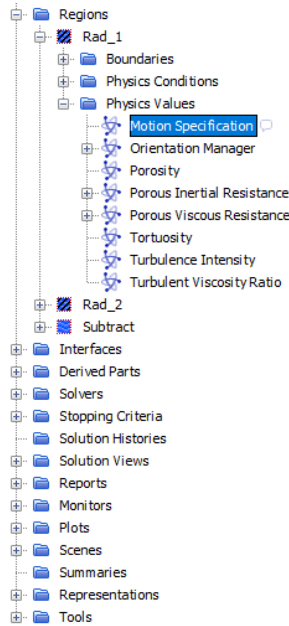


Figure 8: Setting the rotating frame to be the region frame of the first radiator region of the simulation. The same is done to regions Rad_2 and Subtract.

those belonging to the domain of the flow. These must be set to stationary in the laboratory frame. This parameter can be found under Regions > Boundaries > Boundary_Name > Physics Conditions > Reference Frame Specification. This option must be changed for the boundaries corresponding to the inlet, outlet, ground, upper and walls per the CAD section's naming convention. For the upper and walls conditions, these are made slip walls, which mathematically is the same as setting them to being symmetry planes. Either approach is acceptable. The ground is set to be a no-slip wall, but with no specified tangential velocity, since the car is moving, not the air. Follow best meshing procedures for these conditions in terms of prism layer cells. Local

rotation rates are then applied to each wheel, with the motion being centred about each wheel's z-axis. The wall relative velocity should be set to constant, defined by the rotation rate that corresponds to the tire. On the inlet boundary condition, the velocity magnitude should 0 m/s and the pressure outlet should be set to the default conditions.

The final changes before running are to the physics continuum initial conditions. Here, the initial velocity should be set to 0 m/s in every direction. **The simulation is now ready for solving in the usual manner. Simulations can take anywhere from 2000-4000 iterations using the segregated solver, depending on geometry and cell count. Experiment with your own geometry to determine the number of iterations required.**

7 Running and Results

During data collection, residuals will vary from car to car and by meshing methodology. Based on run experience, residuals oscillate around the following regions for a converged solution (residuals are normalized):

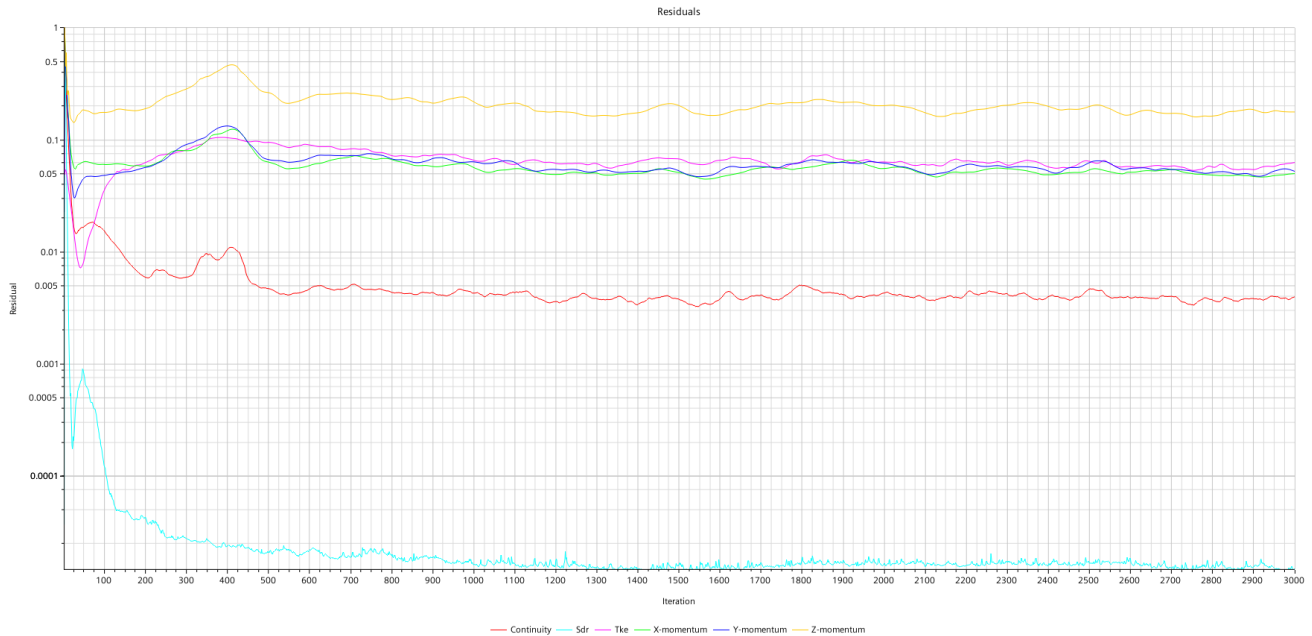


Figure 9: Residual plot for a converged solution for a 0° yaw case, for a 8.375 m corner taken with a lateral acceleration of 1.1 G's.

Some unsteadiness in both residuals and coefficients/forces is to be expected. Even on elements such as front wings, some solution unsteadiness should be expected. In all runs Queen's Formula has done, the data are averaged over the final 1000 iterations of solution time, where the residuals are in their "steady" state. Results for main coefficients are as follows.

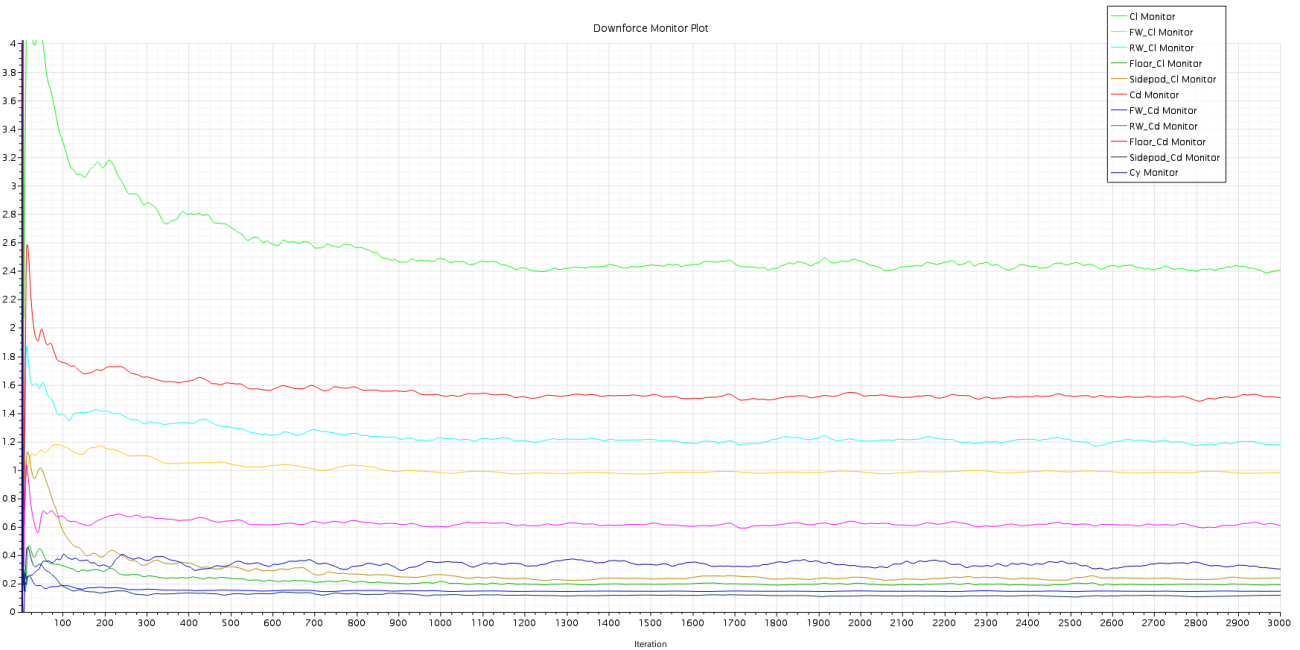


Figure 10: Force coefficients for the same run conditions as the presented residual plot. The force directions are taken with the laboratory coordinates, with downforce being positive and a side force pointing into the turn being positive

In post-processing, if streamlines are to be created, it is important to set the Vector Field option to Velocity in Rotating. This will set the reference frame to be the rotating frame, and will create the desired streamline patterns as presented in Figure 1. Using Velocity as the vector field, will give nonsensical results.

Development of your car from a cornering case may garner some interesting results. The Q20 car was developed from the ground up in this cornering simulation. As such, the loss in downforce is around 4-5% compared to straight line simulations.